Science of the Total Environment 806 (2022) 150566

Contents lists available at ScienceDirect

Science of the Total Environment

journal homepage: www.elsevier.com/locate/scitotenv

Characterization of the genetic structure of *mcr-1* gene among *Escherichia coli* isolates recovered from surface waters and sediments from Ecuador

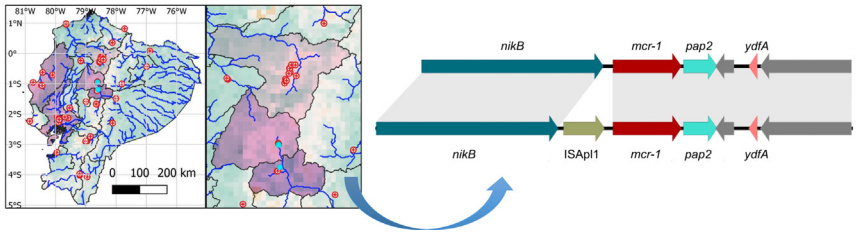
William Calero-Cáceres [a,b,*], Daniel Tadesse [c], Katherine Jaramillo [d], Xavier Villavicencio [d], Efraín Mero [b], Liliana Lalaleo [a], Caitlin Welsh [c], José E. Villacís [d], Emmanuelle Quentin [d], Henry Parra [e], Maria Soledad Ramirez [f], Anthony D. Harries [g,h], José L. Balcázar [i,j]

[a] UTA-RAM-One Health, Department of Food and Biotechnology Science and Engineering, Universidad Técnica de Ambato, Ambato, Ecuador
[b] Centro de Investigaciones Agropecuarias, Facultad de Ciencias Agropecuarias, Universidad Técnica de Ambato, Cevallos, Ecuador
[c] U.S. Food & Drug Administration, Center for Veterinary Medicine, Office of Research Laurel, MD 20708, USA
[d] Centro de Referencia Nacional de Resistencia a los Antimicrobianos RAM, Instituto Nacional de Investigación en Salud Pública "Dr. Leopoldo Izquieta Pérez" INSPI, Quito, Ecuador
[e] Centro de Investigación Microbiológica Dr. Henry Parra, Guayaquil, Ecuador
[f] Department of Biological Science, College of Natural Sciences and Mathematics, California State University Fullerton, Fullerton, CA, USA
[g] International Union Against Tuberculosis and Lung Disease, Paris, France
[h] London School of Hygiene and Tropical Medicine, London, UK
[i] Catalan Institute for Water Research (ICRA), 17003 Girona, Spain
[j] University of Girona, 17004 Girona, Spain

HIGHLIGHTS

- First report of *mcr-1* gene in *E. coli* from environmental samples in Ecuador
- The isolates were obtained from irrigation channels used for agricultural purposes.
- The lack of sewage treatment coverage could favor the spread of resistant bacteria.
- cgMLST showed a great diversity among the evaluated isolates.

GRAPHICAL ABSTRACT



ARTICLE INFO

Article history:
Received 4 August 2021
Received in revised form 2 September 2021
Accepted 20 September 2021
Available online 25 September 2021

Editor: Warish Ahmed

Keywords:
Antibiotic resistance
Mcr-1 gene
Escherichia coli
Environment

ABSTRACT

Although anthropogenic activities contribute to the selection and spread of antibiotic resistance in aquatic environments, limited information is available from countries with absent or incomplete sewage treatment systems and the impact of their discharges onto water bodies. This study therefore aimed to characterize the genetic structure of colistin resistance (*mcr*) genes among *Escherichia coli* isolates recovered from surface waters and sediments in Ecuador. Out of 459 isolates, four *Escherichia coli* showed multidrug-resistant phenotypes, which harbored the *mcr-1* gene and β-lactamases, such as bla_{TEM}, $bla_{CTX\text{-}M\text{-}15}$, $bla_{CTX\text{-}M\text{-}55}$, or $bla_{CTX\text{-}M\text{-}65}$ genes. Three *E. coli* isolates (U20, U30 and U144) shared a similar genetic environment surrounding the *mcr-1* gene, which was located on plasmids. Only one *E. coli* isolate (U175) showed that the *mcr-1* gene was chromosomally located. Moreover, the core genome multilocus sequence typing (cgMLST) analysis revealed that these isolates belong to different lineages. This study represents the first detection of the *mcr-1* gene in multidrug-resistant *E. coli* isolates from environmental samples in Ecuador.

© 2021 Elsevier B.V. All rights reserved.

* Corresponding author at: UTA-RAM-One Health Group, Food and Biotechnology Science and Engineering Department, Los Chasquis y Río Payamino, Ambato, Ecuador.
E-mail address: wr.calero@uta.edu.ec (W. Calero-Cáceres).

https://doi.org/10.1016/j.scitotenv.2021.150566
0048-9697/© 2021 Elsevier B.V. All rights reserved.

1. Introduction

Antibiotic resistance has become a serious and growing global public health threat (Ferri et al., 2017; Hernando-Amado et al., 2019). Its study has traditionally been focused on clinically relevant pathogens; however, the containment of the emergence and spread of antibiotic resistance determinants requires a holistic integration of human, animal and environmental health (Lin et al., 2018; Martinez, 2009). In fact, there is growing evidence of the role of the environment in both the selection and spread of antibiotic-resistant bacteria (ARB) and antibiotic-resistance genes (ARGs), especially in anthropogenically-impacted ecosystems (Drieux et al., 2008; Colomer-Lluch et al., 2014; Marti et al., 2014).

Urban sewage represents one of the main sources of ARB and ARGs (Bouki et al., 2013; Novo et al., 2013; Colomer-Lluch et al., 2014; Barancheshme and Munir, 2018), and their discharges (even treated ones) into water bodies could increase the levels of these emerging pollutants in the environment, whereby disturbing the ecological balance (Berglund, 2015; Maestre-Carballa et al., 2019; Swift et al., 2019). Although several studies have demonstrated that urban discharges promote the persistence and spread of antibiotic resistance, limited information is available from developing countries. Moreover, this impact can be exacerbated in countries, where the coverage of sewage treatment is still scarce (WWAP (United Nations World Water Assessment Programme), 2017).

In the case of Ecuador, there is a lack of adequate sewage treatment facilities (WHO, 2019) and recent studies have demonstrated a high prevalence of ARB in anthropogenically-impacted surface waters in Ecuador, compared with those undisturbed systems (Guerrero-Latorre et al., 2018; Ortega-Paredes et al., 2019a). Considering that colistin is a last resort antibiotic for the treatment of multidrug-resistant Gram-negative bacterial infections (Al-Tawfiq et al., 2017; WHO, 2017), this study aimed to characterize the genetic structure of the *mcr*-1 gene among *E. coli* isolates recovered from surface waters and sediments in Ecuador. A core genome multilocus sequence typing (cgMLST) was also used to assess the relatedness among these isolates.

2. Materials and methods

2.1. Bacterial isolation

Sampling points, which covered the coastal and Andean regions in Ecuador, were selected according to the spatial suitability criteria, such as population density, agricultural areas, river network, irrigation channels, proximity to hospitals, logistic feasibility, and upstream and downstream areas from the discharge point. Fifty-nine water (1.0 L) and sediment (100 g, depth 1 cm) samples were collected from January to October 2018 from rivers and irrigation channels in the provinces of Pichincha ($n = 6$), Cotopaxi ($n = 12$), Tungurahua ($n = 15$), Manabí ($n = 13$), and Guayas ($n = 13$). All samples were collected in sterile containers, transported to the laboratory at 5 °C ± 2 °C, and processed immediately for microbiological analysis and further assays. Briefly, serial dilutions of water and sediment samples, the latter previously homogenized in PBS at pH 7.4 (1:10 w/v), were filtered through 0.45 μm cellulose membrane filters (Merck Millipore; Darmstadt, Germany) and placed on both Chromocult agar according to the ISO 9308 (International Organization for Standardization, 2014) and MacConkey agar (Merck Millipore) (Sigma-Aldrich; St. Louis, USA). After the incubation period at 37 °C for 24 h, blue-violet colonies in Chromocult agar and both lactose-positive and negative colonies in MacConkey agar were purified and cryopreserved with 15% glycerol. Up to 10 colonies from each sampling site were analyzed to avoid the isolation of clonal bacteria.

2.2. PCR detection of mcr genes and bacterial identification

All isolates were screened via PCR to detect the presence of genes conferring resistance to colistin such as *mcr-1*, *mcr-2*, *mcr-3*, *mcr-4*, and *mcr-5*. Moreover, all isolates positive for *mcr* genes were further evaluated for the presence of β-lactamase genes such as bla_{IMI}, bla_{KPC}, bla_{OXA-48}, bla_{NDM}, bla_{VIM}, bla_{TEM}, and bla_{CTX-M} groups. PCR amplification was performed in a total volume of 12.5 μL using DreamTaq Green PCR Mastermix 6.3 μL (Thermo Fisher Scientific; Waltham, MA, USA), 0.3 μL of each primer (10 μM stock), 4.3 μL of nuclease-free water (Thermo Fisher Scientific), and 1.3 μL of total DNA. All primers used in this study are described in Supplementary Table S1. Amplification was carried out as follows: initial denaturation at 95 °C for 1 min; 35 cycles of 95 °C for 30 s, annealing (temperatures given in Supplementary Table S1) for 30 s, extension at 72 °C for 1 min; and a final elongation step at 72 °C for 7 min. An aliquot of 10 μL of each PCR product was analyzed via agarose gel electrophoresis and stained using SYBR® Safe DNA Gel Stain (Invitrogen, Carlsbad, USA). Positive amplicons were then sequenced by Elim Biopharm (Hayward, USA). Moreover, *mcr*-positive isolates were identified using the MALDI Biotyper system (Bruker Daltonik GmbH, Bremen, Germany).

2.3. Whole genome sequencing

The whole genome of all isolates positive for *mcr* genes was sequenced using the Illumina MiSeq platform (Illumina, San Diego, CA, USA). In order to determine the genetic environment of *mcr* genes, the sequences were assembled into contigs using QIAGEN CLC Genomics Workbench version 10.0 (Qiagen, Hilden, Germany) (Bankevich et al., 2012), whose assemblies were screened to identify the contig carrying the *mcr* gene using the BLASTN application (Camacho et al., 2009) with the resistance gene sequence as the query sequence. The identified contigs were then aligned against a previously curated database of complete *Enterobacteriaceae* plasmids (Orlek et al., 2017), as well as against reference genomes using the GenBank database (https://www.ncbi.nlm.nih.gov/genbank/). The results were visualized using the Easyfig program (Sullivan et al., 2011).

2.4. Molecular typing of mcr-positive isolates

Seven loci (*adk*, *fumC*, *gyrB*, *icd*, *mdh*, *purA* and *recA*), previously described for *E. coli*, were used for the initial MLST analysis. A core genome MLST (cgMLST) scheme for *E. coli* was created using Ridom SeqSphere+ software v7.0 (Jünemann et al., 2013), with the *E. coli* K-12 genome as reference. The cgMLST scheme containing 3166 gene targets was used to investigate the relatedness among the isolates and with those *mcr*-positive isolates previously reported in other countries (Supplementary Table S2). Six *mcr*-negative *E. coli* isolates were also included in the analysis, which were previously isolated from clinical samples in Ecuador. Genome assemblies were also uploaded to the Center for Genomic Epidemiology to extract information on ARGs and plasmid replicons for *E. coli* using ResFinder 4.1 and PlasmidFinder 2.1, respectively (Bortolaia et al., 2020; Carattoli et al., 2014).

2.5. Conjugation assays and antibiotic susceptibility testing

Conjugation assays were performed using *E. coli* J53 as the recipient strain, as previously described (Liu et al., 2015). The donor-recipient ratio was 1:1 using nutrient agar supplemented with colistin sulfate (0.5 μg/mL) and sodium azide (100 μg/mL) as the selective medium. The transconjugants were analyzed via the Vitek® 2 Compact system (BioMérieux; Marcy-lEtoile, France) to compare their susceptibility profiles with the donor strain. The Sensititre System with ARGNF plates was used to determine the antimicrobial susceptibility of *E. coli* transconjugants. Minimum inhibitory concentrations of piperacillin/tazobactam (TPZ), ceftazidime (CAZ), cefotaxime (CTX), cefepime (FEP), aztreonam (ATM), ertapenem (ETP), imipenem (IPM), meropenem (MEM), amikacin (AMK), gentamicin (GEN), ciprofloxacin (CIP), Trimethoprim/sulfamethoxazole (SXT) and colistin (COL) were evaluated according to the CLSI guidelines (Clinical and Laboratory

Standards Institute, 2020). Phenotypic β-lactamase production (ESBL) was evaluated via a double-disk synergy test (Drieux et al., 2008). Phenotypic testing for carbapenemase type KPC was performed via the boronic acid combined-disk test (PBA) (Tsakris et al., 2009). An EDTA-disk synergy test was used to identify metallo-β-lactamase-producing strains among the *mcr-1* positive isolates and their transconjugants, as previously described (Yan et al., 2004).

3. Results and discussion

A total of 459 *Enterobacteriaceae*-like-morphology isolates were recovered from different sites and dates. Among them, only 4 isolates carried *mcr* genes as demonstrated by PCR amplification. Moreover, MALDI-TOF analysis revealed that *mcr*-positive isolates are *Escherichia coli*. The details about the origin and source of the isolates are shown in Supplementary Table S3.

Multidrug-resistant phenotypes were detected, showing an extended-spectrum β-lactamases (ESBL) phenotype in all *mcr*-positive *E. coli* isolates (Supplementary Table S4). All *mcr*-positive *E. coli* isolates were classified as non-wild type (NWT) according to their colistin MIC determination (>4 mg/L). Additionally, all the isolates showed ESBL phenotype and resistance to clinically-relevant antibiotics such as cefotaxime, ceftazidime, cefepime and aztreonam (Supplementary Table S4). All *mcr*-positive *E. coli* isolates co-harbored different β-lactamases that include $bla_{\text{TEM-1B}}$, $bla_{\text{CTX-M-15}}$ (U175), $bla_{\text{CTX-M-55}}$ (U20), and $bla_{\text{CTX-M-65}}$ (U30 and U144). In Ecuador, the first reported clinical *E. coli* isolates carrying the *mcr-1* gene co-harbored the $bla_{\text{CTX-M-55}}$ variant (Ortega-Paredes et al., 2016), whereas *E. coli* isolates carrying both *mcr-1* and $bla_{\text{CTX-M-65}}$ genes were detected in humans, chickens and dogs (Loayza-Villa et al., 2019; Ortega-Paredes et al., 2019b). However, no reports have described *E. coli* co-harboring *mcr-1* and $bla_{\text{CTX-M-15}}$ genes in Ecuador to date. Moreover, conjugation assays were successful in three *E. coli* isolates (U20, U30 and U144), and their transconjugants carried the *mcr-1* gene and β-lactamases, indicating the presence of these genes on a plasmid.

Genome analysis revealed that all *mcr* sequences shared 100% nucleotide identity with the sequence of *mcr*-1 that was first found in 2015 in China (Liu et al., 2016). Although the genetic environment of *mcr-1* was similar in 3 (U20, U30 and U144) of out 4 isolates, phenotypic and genotypic antibiotic resistance features were different among these isolates. Moreover, the genetic environment of *mcr-1* in these isolates was different to those described for the first time in China (Fig. 1); however, they were similar to those described in plasmids from *E. coli* isolated from animal and municipal wastewater samples in Japan (Hayashi et al., 2019) and patients in Australia (Ellem et al., 2017). The genetic environment surrounding *mcr-1* showed that the *nikB* gene was present in the upstream region (although IS*Apl1* was found to be absent), whereas the *pap2* and *ydfA* genes were present in the downstream region. Interestingly, the *mcr-1* gene was chromosomally located in one isolate (U175), and the genetic environment, in which two copies of IS*Apl1* is flanking *mcr-1*, was similar to those exclusively described in some *E. coli* chromosomes from Japan and Vietnam (GenBank accession numbers: LC424789, LC424782 and AP021890). These results support our previous observations because transconjugation assays failed in *E. coli* U175.

The cgMLST analysis showed that the *E. coli* isolates belong to different lineages (Fig. 2), which include the ST10 (isolate U175) that has been detected in *mcr*-positive *E. coli* of animal origin from Spain (Migura-Garcia et al., 2020) and clinical samples from Colombia (Saavedra et al., 2017) and China (Huang et al., 2020). Moreover, the ST5876 (isolate U144), commonly associated with animal origin (Zhou et al., 2020), has been detected in *mcr*-positive *E. coli* in Japan, the Netherlands and Spain (Fig. 2). The ST58 (isolate U30) belonging to the ST155 complex has recently been associated with *mcr*-positive *E. coli* from a wide variety of sources (animal, human, environmental). However, no reports of *E. coli* belonging to ST10779 (isolate U20) have been found to harbor *mcr* genes. Likewise, all the isolates harbored a wide diversity of aminoglycoside, β-lactam, phenicol and tetracycline resistance genes (Table 1), whose occurrence has usually been associated with isolates harboring the *mcr*-1 gene in samples of human, animal and food origin (Hammad et al., 2019; Kassem et al., 2021; Migura-Garcia et al., 2020). Moreover, in silico plasmid replicon typing revealed that *mcr*-positive *E. coli* isolates harbored multiple plasmid replicons (Table 1).

Due to the limited number of *mcr*-positive *E. coli* is impossible to find an epidemiological connection between the lack of adequate sewage treatment, and their subsequent discharges into water bodies, and the occurrence of these isolates in water bodies in Ecuador. However, untreated sewage discharges represent a serious risk in terms of food and water safety. The Cutuchi River receives untreated urban sewage from the city of Latacunga and rural parishes, thereby estimating their impact in 30,000 m^3 of sewage/day and 18 tons/day of garbage and waste from domestic origin, unofficial dumps, hospitals, dead animals, and industrial sewage (GAD Latacunga, 2014). Currently, the council of Latacunga is constructing a wastewater treatment plant that will provide wastewater treatment for 170,000 inhabitants, according to the last population census (GAD Latacunga, 2021; INEC, 2010). However, considering the lack of adequate sanitary facilities in other points of the basin, the potential effect of this construction and future operation would not be enough to reduce the presence of these biological contaminants.

In addition, the Latacunga-Salcedo-Ambato irrigation channel receives its main supply from the Cutuchi River, which is impacted by untreated urban sewage discharges from Latacunga city and their surrounding rural areas. It should be noted that this water supply is used for agricultural purposes and there is no evidence about its

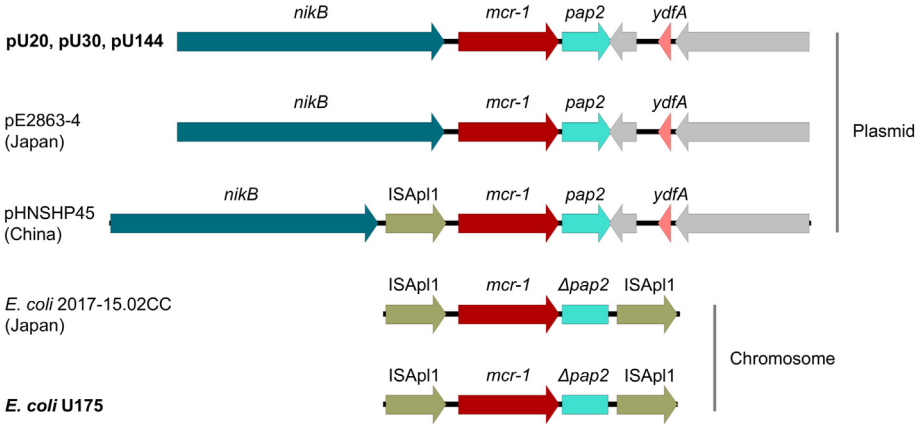


Fig. 1. Genetic environment of the *mcr-1* gene in *mcr*-positive *E. coli* isolates. The open reading frames (ORFs) are illustrated by arrows pointing toward their respective orientation.

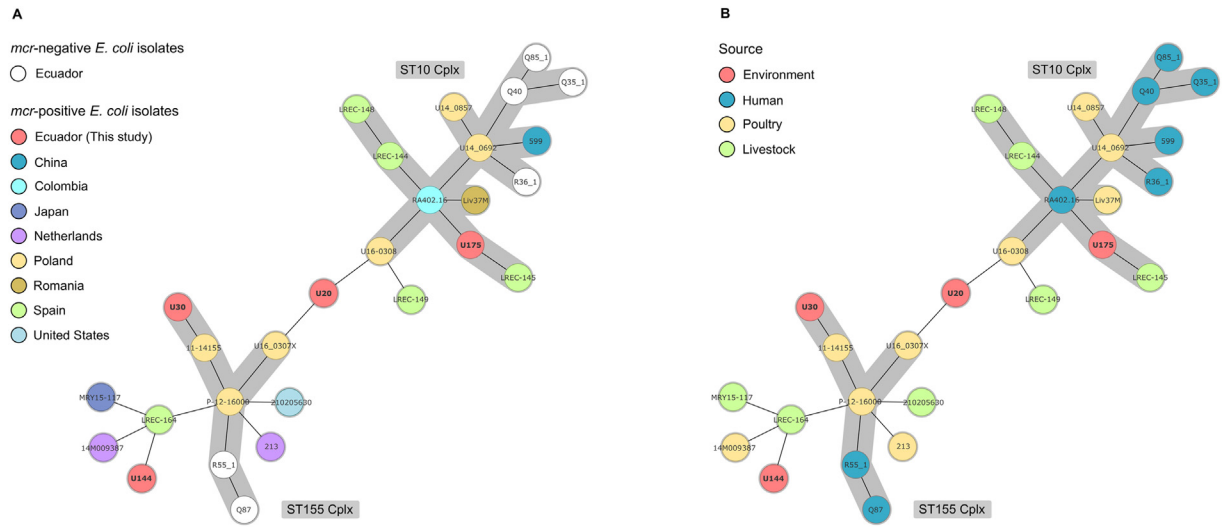


Fig. 2. Minimum spanning tree of the core-genome multilocus sequence type allelic profiles of *mcr*-positive *E. coli* isolates from this study and those obtained from previous studies. Clusters are colored and labeled according to the country of origin (A) and the source of isolation (B).

influence on the sanitary quality of surrounding areas. It is necessary therefore to establish an adequate monitoring system that evaluates deeply their impact on the environment and on the influenced population.

4. Conclusions

In this work, we reported for the first time the presence of multidrug-resistant *E. coli* harboring the *mcr-1* gene from water bodies and their sediments in Ecuador. This suggests the circulation of *mcr*-positive *E. coli* isolates in the environment, which could have an anthropogenic origin, including urban wastewater discharges and livestock facilities. These findings pose a potential threat to public health, considering both that the isolates were obtained from irrigation channels used for agricultural purposes and that the isolates showed multidrug-resistant phenotypes. Finally, this study puts in evidence the necessity to take suitable measures that minimize the spread of multidrug-resistant bacteria in water bodies in Ecuador.

Supplementary data to this article can be found online at https://doi.org/10.1016/j.scitotenv.2021.150566.

CRediT authorship contribution statement

W.C-C., J.E.V. and J.L.B. designed the investigation. E.Q. performed the spatial suitability criteria for the sampling points. K.J., X.V., E.M., and L.L. collected the samples and performed the preliminary screening for *mcr*-positive bacteria. H.P. performed bacterial identification through MALDI-TOF. D.T. and C.W. performed the Illumina Sequencing and preliminary analyses of the data. K.J. and X.V. performed antimicrobial susceptibility testing, conjugation assays and phenotypic synergies. W.C-C., and J.L.B. designed and performed genomic analysis, including genome assemblies, MLST, antibiotic resistance genes, plasmid and cgMLST analysis. M.S.R. and A.D.H. collaborated in the protocol designing according to Structured Operational Research and Training Initiative SORT IT (Partnership between TDR-UNICEF, UNDP, World Bank and WHO). W.C-C., A.D.H., and M.S.R. obtained the approbation from the Ethics Advisory Groups from PAHO and The Union. W.C-C. supervised the entire work. W.C-C. and J.L.B. wrote the manuscript. All authors contributed to manuscript revision and editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors would thank to David Mendoza, Xiomara Garcés and Vanessa Sailema for their excellent technical assistance. J.L. Balcázar acknowledges the support from the Economy and Knowledge Department of the Catalan Government through Consolidated Research Group (ICRA-ENV 2017 SGR 1124) and from the CERCA Programme. The views expressed in this article are those of the authors and do not necessarily reflect the official policy of the Department of Health and Human Services, the U.S. Food and Drug Administration, or the U.S.

Table 1
Whole-genome sequencing-based characterization of four *mcr*-positive *E. coli* isolates from surface waters and sediments from Ecuador.

Isolate	Source	Origin	No. of contigs	Total length (bp)	N50 (bp)	MLST	Plasmid type	Antibiotic resistance genes	BioSample accession no.
U175	Irrigation channel (sediment)	Latacunga, Cotopaxi	202	4,976,364	88,935	ST10	Col, IncFIA, IncFIB, IncFII, IncR, and IncX1	*aac(3)-IIa*, *aadA1*, *aadA2*, *strA*, *strB*, *mcr-1*, *sul2*, *sul3*, *dfrA14*, *tet(A)*, *tet(M)*, *cmlA1*, *floR*, *bla*CTX-M-15, *bla*TEM-1, and *qnrS1*	SAMN20560763
U144	River (sediment)	Latacunga, Cotopaxi	104	5,376,946	210,505	ST5876	IncFII, IncI, Inc2, and IncR	*aac(3)-IVa*, *aph(4)-Ia*, *strA*, *strB*, *mcr-1*, *tet(B)*, *floR*, *bla*CTX-M-65, and *bla*TEM-1	SAMN20560762
U30	River (water)	Latacunga, Cotopaxi	154	5,156,418	110,489	ST58	Col, IncFIB, IncFII, IncI, IncI2, and IncQ1	*aac(3)-IVa*, *aph(3′)-Ia*, *aph(4)-Ia*, *strA*, *strB*, *mcr-1*, *sul2*, *dfrA5*, *tet(A)*, *floR*, *bla*CTX-M-65, *bla*TEM-1, *qnrB19*, and *fosA*	SAMN20560761
U20	Irrigation channel (water)	Atahualpa, Tungurahua	189	4,912,376	75,080	ST10779	IncFIB, IncFIC, IncFII, and IncI2	*aadA1*, *aadA2*, *mcr-1*, *sul3*, *dfrA1*, *tet(B)*, *cmlA1*, *bla*CTX-M-55, *bla*TEM-1, and *fosA*	SAMN20560760

government. Reference to any commercial materials, equipment, or process does not in any way constitute approval, endorsement, or recommendation by the Food and Drug Administration. This article was developed through the Structured Operational Research and Training Initiative (SORT IT), a global partnership coordinated by TDR, the UNICEF, UNDP, World Bank, WHO Special Programme for Research and Training in Tropical Diseases hosted at the World Health Organization. The specific SORT IT program that led to this research article included an implementation partnership of TDR and the Pan American Health Organization (PAHO), The WHO Country offices of Colombia and Ecuador; the Ministry of Health and Social Protection, Colombia; Food and Agriculture Organization, Sierra Leone; Sustainable Health Systems, Freetown, Sierra Leone; The Tuberculosis Research and Prevention Center Non-Governmental Organization, Armenia; The International Union Against Tuberculosis and Lung Diseases, Paris, France and South East Asia offices, India; Institute of Tropical Medicine, Antwerp, Belgium; Damien Foundation, Belgium; Indian Council of Medical Research- National Institute of Epidemiology; Jawaharlal Institute of Postgraduate Medical Education & Research (JIPMER); GMERS Medical College Gotri Vadodara Gujarat, India; India Medical College Baroda, Gujarat, India; Sri Manakula Vinayagar Medical College, India; Public Health, Ontario; The Universidade Federal de Ciencias de Saude de Porto Alegre, Brazil; Universidade de Brasilia, Brazil; Universidad de Concepcion, Chile, Universidad de los Andes, Colombia; Universidad Pontificia Bolivariana, Colombia; Universidad Pedagógica y Tecnológica de Colombia; The Central University of Ecuador and; California State University of Fullerton, USA; The Autonomous University of Yucatán, México.

Funding section

This study was supported by the Directorate of Research-Universidad Técnica de Ambato (Project SSFCIAL01), the Instituto Nacional de Salud Pública (INSPI), the Food and Drug Administration. The SORT IT AMR Programme is funded by the National Institute of Health Research, Department of Health & Social Care of the United Kingdom and supported by implementing partners. All open access and ethics related costs will be covered by TDR.

References

Al-Tawfiq, J.A., Laxminarayan, R., Mendelson, M., 2017. How should we respond to the emergence of plasmid-mediated colistin resistance in humans and animals? Int. J. Infect. Dis. 54, 77–84. https://doi.org/10.1016/j.ijid.2016.11.415.

Bankevich, A., Nurk, S., Antipov, D., Gurevich, A.A., Dvorkin, M., Kulikov, A.S., Lesin, V.M., Nikolenko, S.I., Pham, S., Prjibelski, A.D., Pyshkin, A.V., Sirotkin, A.V., Vyahhi, N., Tesler, G., Alekseyev, M.A., Pevzner, P.A., 2012. SPAdes: a new genome assembly algorithm and its applications to single-cell sequencing. J. Comput. Biol. 19, 455–477. https://doi.org/10.1089/cmb.2012.0021.

Barancheshme, F., Munir, M., 2018. Strategies to combat antibiotic resistance in the wastewater treatment plants. Front. Microbiol. https://doi.org/10.3389/fmicb.2017.02603.

Berglund, B., 2015. Environmental Dissemination of Antibiotic Resistance Genes and Correlation to Anthropogenic Contamination With Antibiotics, pp. 1–10.

Bortolaia, V., Kaas, R.S., Ruppe, E., Roberts, M.C., Schwarz, S., Cattoir, V., Philippon, A., Allesoe, R.L., Rebelo, A.R., Florensa, A.F., Fagelhauer, L., Chakraborty, T., Neumann, B., Werner, G., Bender, J.K., Stingl, K., Nguyen, M., Coppens, J., Xavier, B.B., Malhotra-Kumar, S., Westh, H., Pinholt, M., Anjum, M., Duggett, N.A., Kempf, I., Nykäsenoja, S., Olkkola, S., Wieczorek, K., Amaro, A., Clemente, L., Mossong, J., Losch, S., Ragimbeau, C., Lund, O., Aarestrup, F.M., 2020. ResFinder 4.0 for predictions of phenotypes from genotypes. J. Antimicrob. Chemother. 75 (12), 3491–3500. https://doi.org/10.1093/jac/dkaa345.

Bouki, C., Venieri, D., Diamadopoulos, E., 2013. Detection and fate of antibiotic resistant bacteria in wastewater treatment plants: a review. Ecotoxicol. Environ. Saf. 91, 1–9. https://doi.org/10.1016/j.ecoenv.2013.01.016.

Camacho, C., Coulouris, G., Avagyan, V., Ma, N., Papadopoulos, J., Bealer, K., Madden, T.L., 2009. BLAST+: architecture and applications. BMC Bioinformatics 10 (1), 1–9. https://doi.org/10.1186/1471-2105-10-421.

Carattoli, A., Zankari, E., García-Fernández, A., Voldby Larsen, M., Lund, O., Villa, L., Møller Aarestrup, F., Hasman, H., 2014. In silico detection and typing of plasmids using PlasmidFinder and plasmid multilocus sequence typing. Antimicrob. Agents Chemother. 58 (7), 3895–3903. https://doi.org/10.1128/AAC.02412-14.

Clinical and Laboratory Standars Institute, 2020. CLSI M100 ED30 Performance Standards for Antimicrobial Susceptibility Testing.

Colomer-Lluch, M., Calero-Cáceres, W., Jebri, S., Hmaied, F., Muniesa, M., Jofre, J., 2014. Antibiotic resistance genes in bacterial and bacteriophage fractions of tunisian and spanish wastewaters as markers to compare the antibiotic resistance patterns in each population. Environ. Int. 73, 167–175. https://doi.org/10.1016/j.envint.2014.07.003.

Drieux, L., Brossier, F., Sougakoff, W., Jarlier, V., 2008. Phenotypic detection of extended-spectrum β-lactamase production in enterobacteriaceae: review and bench guide. Clin. Microbiol. Infect. 14, 90–103. https://doi.org/10.1111/j.1469-0691.2007.01846.x.

Ellem, J.A., Ginn, A.N., Chen, S.C.A., Ferguson, J., Partridge, S.R., Iredell, J.R., 2017. Locally acquired mcr-1 in Escherichia coli, Australia, 2011 and 2013. Emerg. Infect. Dis. 23, 1160–1163. https://doi.org/10.3201/eid2307.161638.

Ferri, M., Ranucci, E., Romagnoli, P., Giaccone, V., 2017. Antimicrobial resistance: a global emerging threat to public health systems. Crit. Rev. Food Sci. Nutr. 57 (13), 2857–2876. https://doi.org/10.1080/10408398.2015.1077192.

GAD Latacunga, 2021. Plan Maestro de Alcantarillado Latacunga [WWW document]. https://www.latacunga.gob.ec/es/plan-maestro-de-alcantarillado.html#.

Guerrero-Latorre, L., Romero, B., Bonifaz, E., Timoneda, N., Rusiñol, M., Girones, R., Rios-Touma, B., 2018. Quito's virome: metagenomic analysis of viral diversity in urban streams of Ecuador's capital city. Sci. Total Environ. 645, 1334–1343. https://doi.org/10.1016/J.SCITOTENV.2018.07.213.

Hammad, A.M., Hoffmann, M., Gonzalez-Escalona, N., Abbas, N.H., Yao, K., Koenig, S., Allué-Guardia, A., Eppinger, M., 2019. Genomic features of colistin resistant Escherichia coli ST69 strain harboring mcr-1 on IncHI2 plasmid from raw milk cheese in Egypt. Infect. Genet. Evol. 73, 126–131. https://doi.org/10.1016/j.meegid.2019.04.021.

Hayashi, W., Tanaka, H., Taniguchi, Y., Iimura, M., Soga, E., Kubo, R., Matsuo, N., Kawamura, K., Arakawa, Y., Nagano, Y., Nagano, N., 2019. Acquisition of mcr-1 and cocarriage of virulence genes in avian pathogenic Escherichia coli isolates from municipal wastewater influents in Japan. Appl. Environ. Microbiol. 85. https://doi.org/10.1128/AEM.01661-19.

Hernando-Amado, S., Coque, T.M., Baquero, F., Martínez, J.L., 2019. Defining and combating antibiotic resistance from One Health and Global Health perspectives. Nat. Microbiol. 4 (9), 1432–1442. https://doi.org/10.1038/s41564-019-0503-9.

Huang, H., Dong, N., Shu, L., Lu, J., Sun, Q., Chan, E.W.C., Chen, S., Zhang, R., 2020. Colistin-resistance gene mcr in clinical carbapenem-resistant enterobacteriaceae strains in China, 2014–2019. Emerg. Microbes Infect. 9, 237–245. https://doi.org/10.1080/22221751.2020.1717380.

INEC, 2010. Resultados del censo 2010 de población y vivienda en el Ecuador.

International Organization for Standardization, 2014. ISO 9308-1:2014 - Water Quality – Enumeration of Escherichia coli and Coliform bacteria – Part 1: Membrane Filtration Method for Waters With Low Bacterial Background Flora.

Jünemann, S., Sedlazeck, F.J., Prior, K., Albersmeier, A., John, U., Kalinowski, J., Mellmann, A., Goesmann, A., von Haeseler, A., Stoye, J., Harmsen, D., 2013. Updating benchtop sequencing performance comparison. Nat. Biotechnol. 31 (4), 294–296. https://doi.org/10.1038/nbt.2522.

Kassem, I.I., Mann, D., Li, S., Deng, X., 2021. Draft genome sequences and resistome analysis of multidrug-resistant and mcr-1-harboring E. coli isolated from preharvest poultry in Lebanon. J. Glob. Antimicrob. Resist. 25, 114–116. https://doi.org/10.1016/j.jgar.2021.03.001.

Latacunga, G.A.D., 2014. Ordenanza Para la descontaminación y protección de los ríos y afluentes hídricos del cantón Latacunga.

Lin, C., Franceschi, S., Clifford, G.M., 2018. Why the antibiotic resistance crisis requires a one health approach. Lancet Infect. Dis. 18, 198–206. https://doi.org/10.1016/S1473-3099(17)30653-9.

Liu, C., Qin, S., Xu, H., Xu, L., Zhao, D., Liu, X., Lang, S., Feng, X., Liu, H.M., 2015. New Delhi Metallo-β-Lactamase 1(NDM-1), the dominant carbapenemase detected in carbapenem-resistant Enterobacter cloacae from Henan Province, China. PLoS One. https://doi.org/10.1371/journal.pone.0135044.

Liu, Y.Y., Wang, Y., Walsh, T.R., Yi, L.X., Zhang, R., Spencer, J., Doi, Y., Tian, G., Dong, B., Huang, X., Yu, L.F., Gu, D., Ren, H., Chen, X., Lv, L., He, D., Zhou, H., Liang, Z., Liu, J.H., Shen, J., 2016. Emergence of plasmid-mediated colistin resistance mechanism MCR-1 in animals and human beings in China: a microbiological and molecular biological study. Lancet Infect. Dis. 16, 161–168. https://doi.org/10.1016/S1473-3099(15)00424-7.

Loayza-Villa, M.F., Salinas, L., Tijet, N., Villavicencio, F., Trujillo, R., Salas, S., Rivera, R., Villacís, J., Satan, C., Uchina, L., Muñoz, O., Zurita, J., Melano, R., Reyes, J., Trueba, G., 2019. Diverse Escherichia coli isolates from domestic animal in the same ecuadorian household all carry the colistin resistance gene mcr1.1. J. Glob. Antimicrob. Resist. https://doi.org/10.1016/j.jgar.2019.12.002.

Maestre-Carballa, L., Lluesma Gomez, M., Angla Navarro, A., Garcia-Heredia, I., Martinez-Hernandez, F., Martinez-Garcia, M., 2019. Insights into the antibiotic resistance dissemination in a wastewater effluent microbiome: bacteria, viruses and vesicles matter. Environ. Microbiol. 21, 4582–4596. https://doi.org/10.1111/1462-2920.14758.

Marti, E., Variatza, E., Balcazar, J.L., 2014. The role of aquatic ecosystems as reservoirs of antibiotic resistance. Trends Microbiol. 22 (1), 36–41. https://doi.org/10.1016/j.tim.2013.11.001.

Martinez, J.L., 2009. The role of natural environments in the evolution of resistance traits in pathogenic bacteria. Proc. Biol. Sci. 276, 2521–2530. https://doi.org/10.1098/rspb.2009.0320.

Migura-Garcia, L., González-López, J.J., Martinez-Urtaza, J., Aguirre Sánchez, J.R., Moreno-Mingorance, A., Perez de Rozas, A., Höfle, U., Ramiro, Y., Gonzalez-Escalona, N., 2020. Mcr-colistin resistance genes mobilized by inc X4, IncHI2, and IncI2 plasmids in Escherichia coli of pigs and white stork in Spain. Front. Microbiol. 10, 3072. https://doi.org/10.3389/fmicb.2019.03072.

Novo, A., André, S., Viana, P., Nunes, O.C., Manaia, C.M., 2013. Antibiotic resistance, antimicrobial residues and bacterial community composition in urban wastewater. Water Res. 47, 1875–1887. https://doi.org/10.1016/j.watres.2013.01.010.

Orlek, A., Phan, H., Sheppard, A.E., Doumith, M., Ellington, M., Peto, T., Crook, D., Walker, A.S., Woodford, N., Anjum, M.F., Stoesser, N., 2017. A curated dataset of complete Enterobacteriaceae plasmids compiled from the NCBI nucleotide database. Data in Brief 12, 423–426. https://doi.org/10.1016/j.dib.2017.04.024.

Ortega-Paredes, D., Barba, P., Zurita, J., 2016. Colistin-resistant Escherichia coli clinical isolate harbouring the mcr-1 gene in Ecuador. Epidemiol. Infect. 1–4. https://doi.org/10.1017/S0950268816001369.

Ortega-Paredes, D., Barba, P., Mena-López, S., Espinel, N., Crespo, V., Zurita, J., 2019a. High quantities of multidrug-resistant Escherichia coli are present in the Machángara urban river in Quito, Ecuador. J. Water Health. https://doi.org/10.2166/wh.2019.195.

Ortega-Paredes, D., Haro, M., Leoro-Garzón, P., Barba, P., Loaiza, K., Mora, F., Fors, M., Vinueza-Burgos, C., Fernández-Moreira, E., 2019b. Multi-drug resistant Escherichia coli isolated from canine feces in a public park in Quito, Ecuador. J. Glob. Antimicrob. Resist. https://doi.org/10.1016/J.JGAR.2019.04.002.

Saavedra, S.Y., Diaz, L., Wiesner, M., Correa, A., Arévalo, S.A., Reyes, J., Hidalgo, A.M., de la Cadena, E., Perenguez, M., Montaño, L.A., Ardila, J., Ríos, R., Ovalle, M.V., Díaz, P., Porras, P., Villegas, M.V., Arias, C.A., Beltrán, M., Duarte, C., 2017. Genomic and molecular characterization of clinical isolates of enterobacteriaceae harboring mcr-1 in colombia, 2002 to 2016. Antimicrob. Agents Chemother. 61 (12), e00841-17. https://doi.org/10.1128/AAC.00841-17.

Sullivan, M.J., Petty, N.K., Beatson, S.A., 2011. Easyfig: a genome comparison visualizer. Bioinformatics 27, 1009–1010. https://doi.org/10.1093/bioinformatics/btr039.

Swift, B.M.C., Bennett, M., Waller, K., Dodd, C., Murray, A., Gomes, R.L., Humphreys, B., Hobman, J.L., Jones, M.A., Whitlock, S.E., Mitchell, L.J., Lennon, R.J., Arnold, K.E., 2019. Anthropogenic environmental drivers of antimicrobial resistance in wildlife. Sci. Total Environ. 649, 12–20. https://doi.org/10.1016/j.scitotenv.2018.08.180.

Tsakris, A., Kristo, I., Poulou, A., Themeli-Digalaki, K., Ikonomidis, A., Petropoulou, D., Pournaras, S., Sofianou, D., 2009. Evaluation of boronic acid disk tests for differentiating KPC-possessing Klebsiella pneumoniae isolates in the clinical laboratory. J. Clin. Microbiol. 47, 362–367. https://doi.org/10.1128/JCM.01922-08.

WHO, 2017. WHO updates Essential Medicines List with new advice on use of antibiotics, and adds medicines for hepatitis C, HIV, tuberculosis and cancer. [WWW Document]. World Heal. Organ. http://www.who.int/news-room/detail/06-06-2017-who-updates-essential-medicines-list-with-new-advice-on-use-of-antibiotics-and-adds-medicines-for-hepatitis-c-hiv-tuberculosis-and-cancer.

WHO, 2019. UN-Water Global Analysis and Assessment of Sanitation and Drinking-Water (GLAAS) 2019 Report National Systems to Support Drinking-water, Sanitation and Hygiene - Global Status Report 2019. Geneva.

WWAP (United Nations World Water Assessment Programme), 2017. The United Nations World Water Development Report 2017. Wastewater: The Untapped Resource. UNESCO, Paris.

Yan, J.-J., Wu, J.-J., Tsai, S.-H., Chuang, C.-L., 2004. Comparison of the double-disk, combined disk, and etest methods for detecting metallo-β-lactamases in gram-negative bacilli. Diagn. Microbiol. Infect. Dis. 49, 5–11. https://doi.org/10.1016/j.diagmicrobio.2004.01.002.

Zhou, Z., Alikhan, N.F., Mohamed, K., Fan, Y., Achtman, M., 2020. The EnteroBase user's guide, with case studies on salmonella transmissions, Yersinia pestis phylogeny, and Escherichia core genomic diversity. Genome Res. 30, 138–152. https://doi.org/10.1101/gr.251678.119.